S3 STORAGE CLASSES

Definition

S3 Storage Classes are different categories of storage designed to optimize cost and performance based on how frequently data is accessed and how quickly it needs to be retrieved.

Key Characteristics

Storage classes apply only at the object level, not at the bucket level

Selection depends on:

- Data access pattern (frequent or infrequent)
- Resiliency requirements (multi-AZ or single-AZ)
- Retrieval requirements (instant, minutes, or hours)

Why Storage Classes Are Used

- To reduce storage cost
- To store data based on access frequency
- To meet performance and availability requirements
- To manage long-term data retention and backups

Types of S3 Storage Classes

1. S3 Standard
   Used for frequently accessed data with low latency and high availability.
2. S3 Intelligent-Tiering
   Automatically moves data between access tiers based on usage patterns.
3. S3 Standard-IA (Infrequent Access)
   Used for data accessed less frequently but requires rapid access when needed.
4. S3 One Zone-IA
   Stores data in a single Availability Zone at lower cost for non-critical data.
5. S3 Glacier Instant Retrieval
   Low-cost storage with instant access for archive data (milliseconds).
6. S3 Glacier Flexible Retrieval
   Archive storage with retrieval times ranging from minutes to hours.
7. S3 Glacier Deep Archive
   Lowest-cost storage for long-term data archival with longer retrieval time of 12 to 48 hours

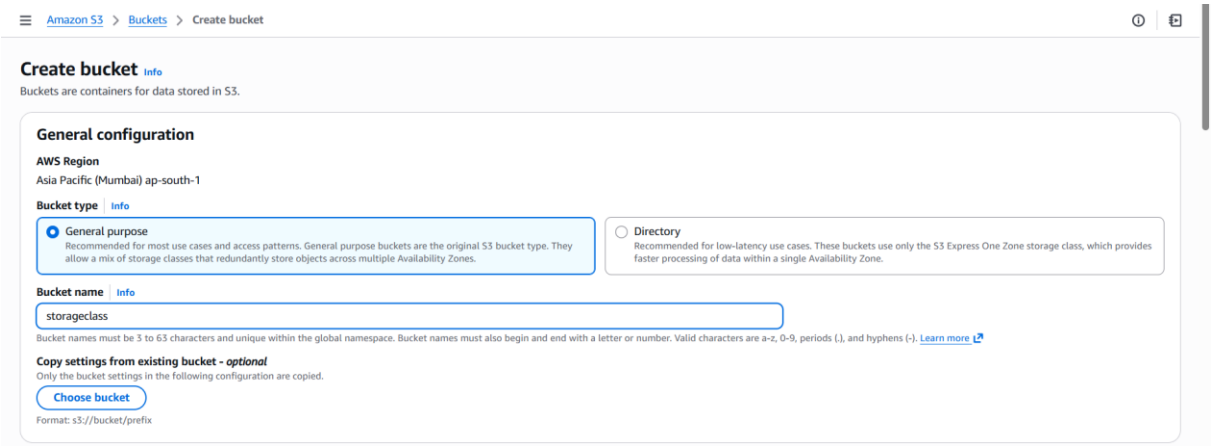
Advantages of Storage Classes

- Cost optimization based on usage
- Flexible data access options
- Suitable for backup, archival, and compliance
- Supports lifecycle policies for automatic transitions

HANDS-ON FOR STORAGE CLASSES AND LIFECYCLE RULE

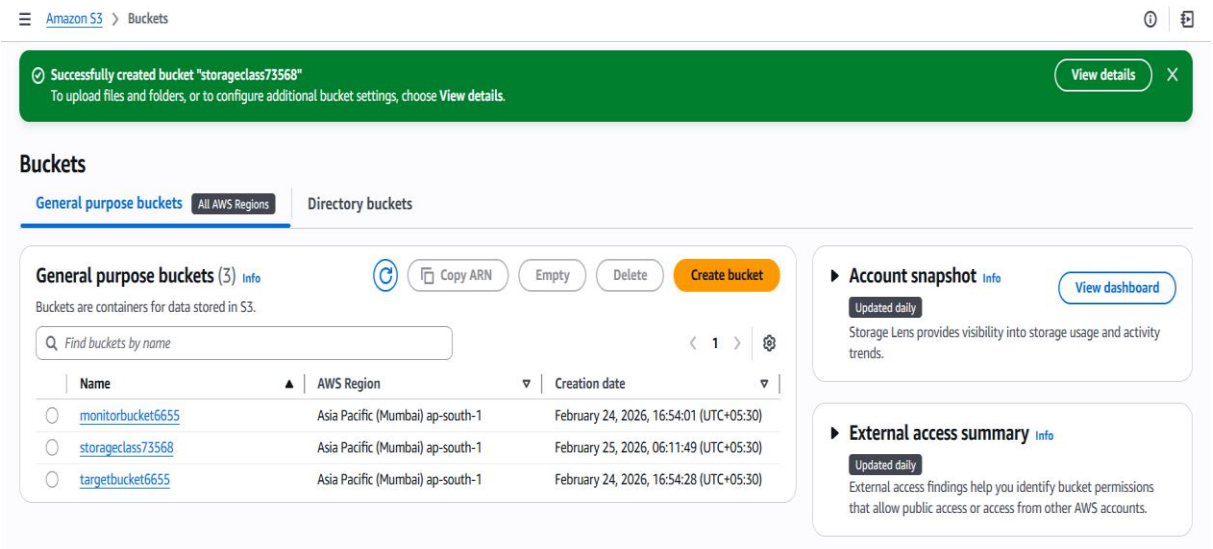
Screenshot 1: Create Bucket and Enter Name

1. Open the AWS S3 console.
2. Click Create bucket.
3. Enter a unique bucket name.
4. Select the required AWS Region.



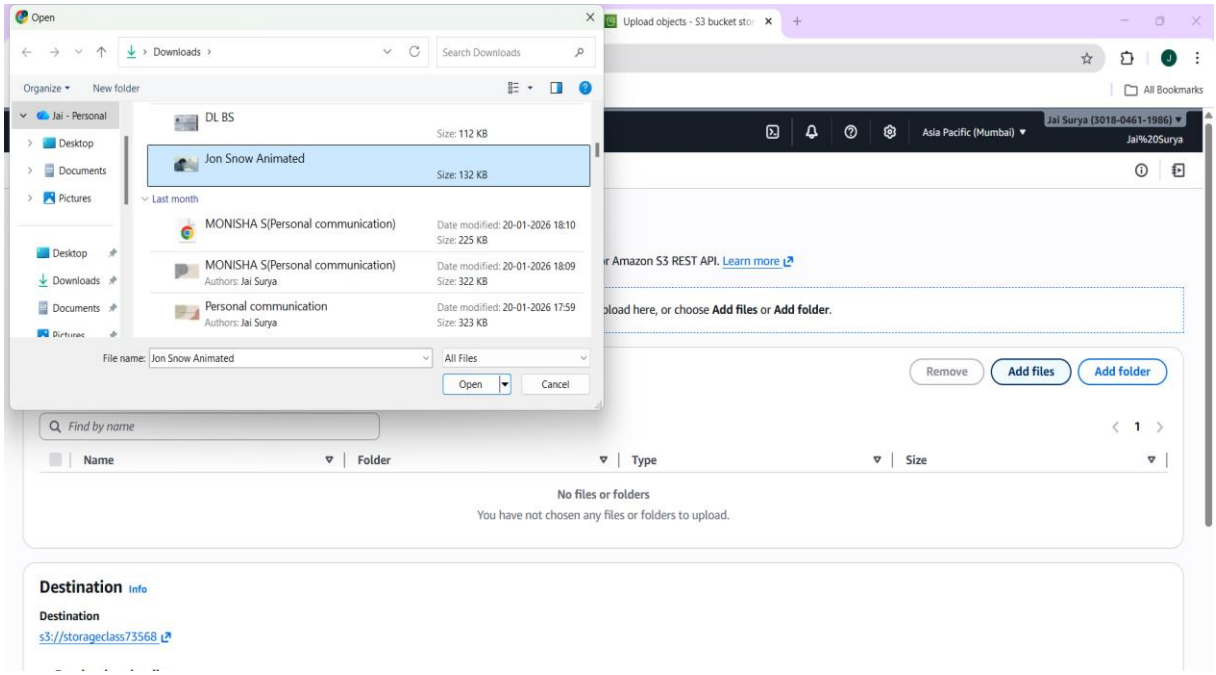
Screenshot 2: Bucket Created Successfully

1. Click Create bucket.
2. The S3 bucket is created successfully.
3. The bucket appears in the S3 bucket list and is ready for use.



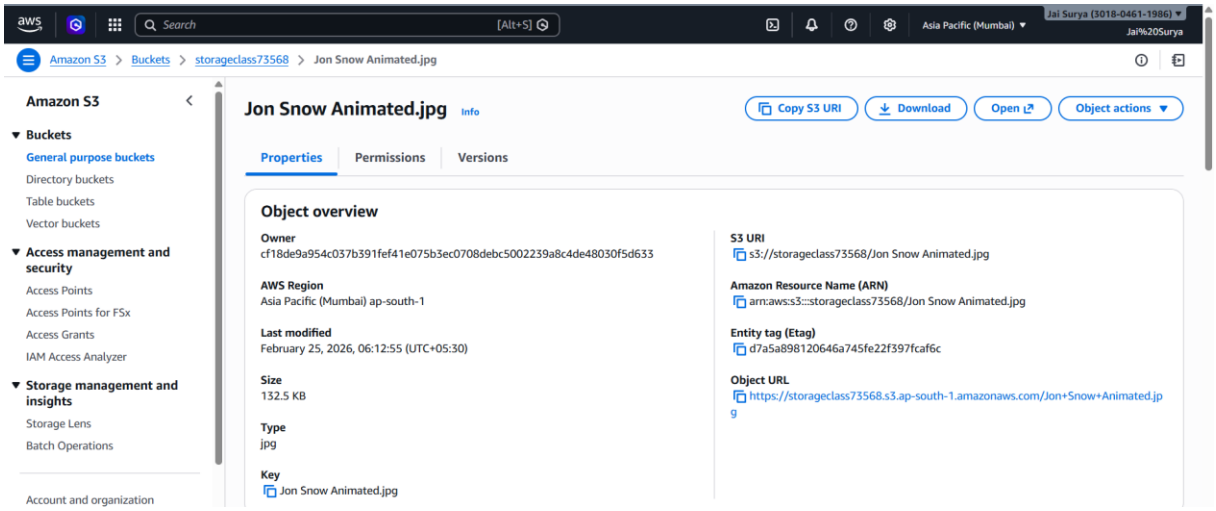
Screenshot 3: Choose File to Upload

1. Open the created S3 bucket.
2. Click Upload.
3. Click Add files.
4. Choose a file from your local system to upload to the bucket.



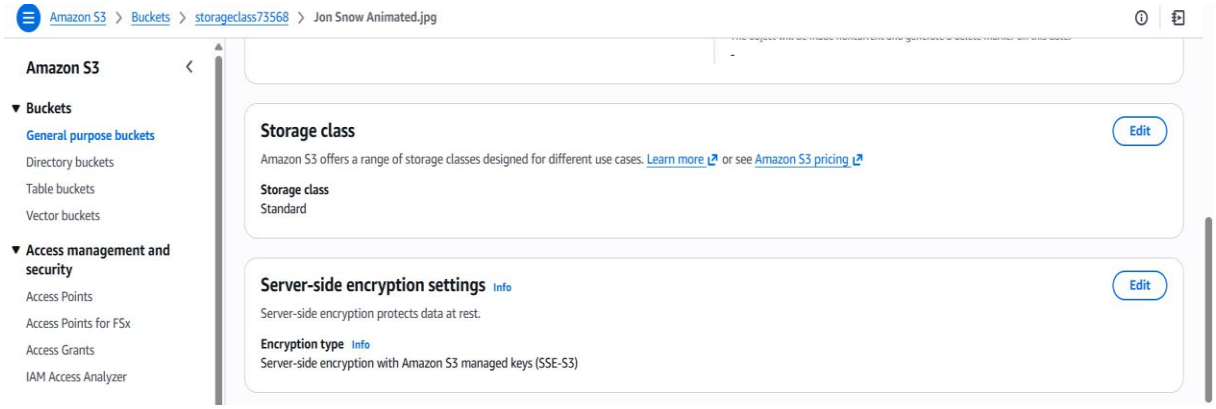
Screenshot 4: View Object Properties

1. After the file is uploaded, select the object in the bucket.
2. Click on the Properties tab.



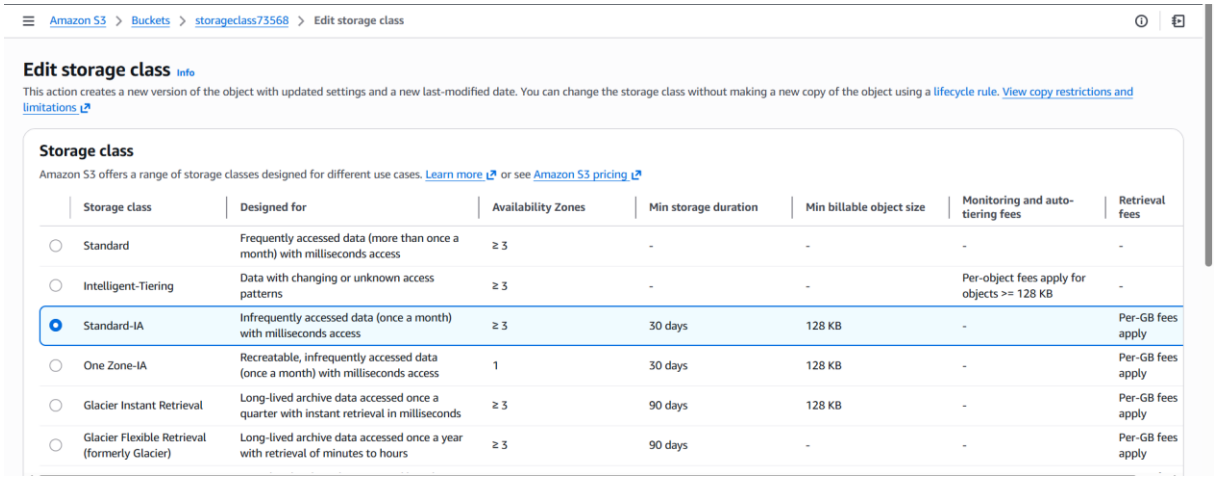
Screenshot 5: Edit Object Storage Class

1. Scroll to the Storage class section.
2. Click Edit to modify the storage class of the object.



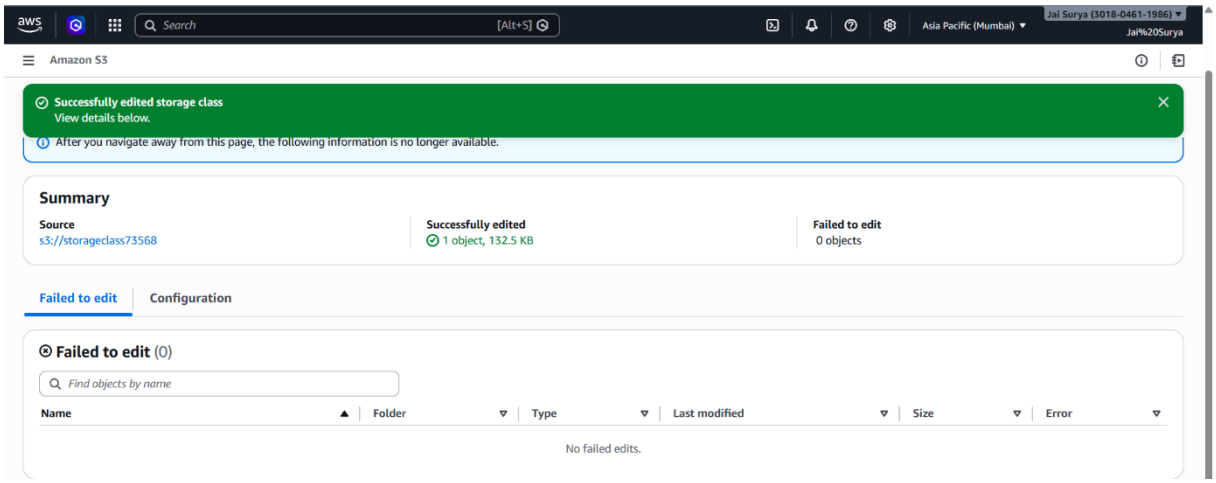
Screenshot 6: Choose Storage Class and Confirm

1. Select the required S3 storage class for the object.
2. Click Save changes to apply the selected storage class.



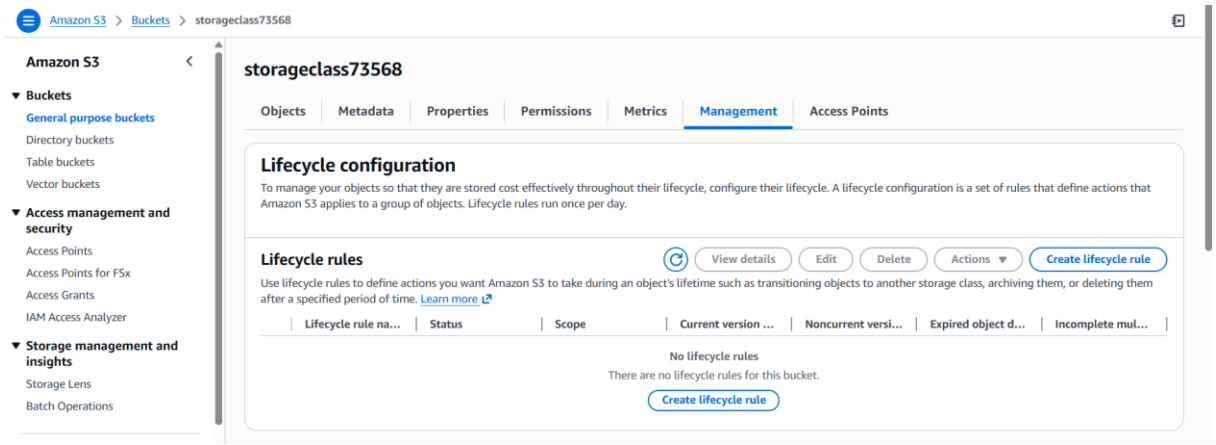
Screenshot 7: Storage Class Updated Successfully

1. The selected storage class is applied successfully to the object.
2. The updated storage class is visible in the Object Properties section.



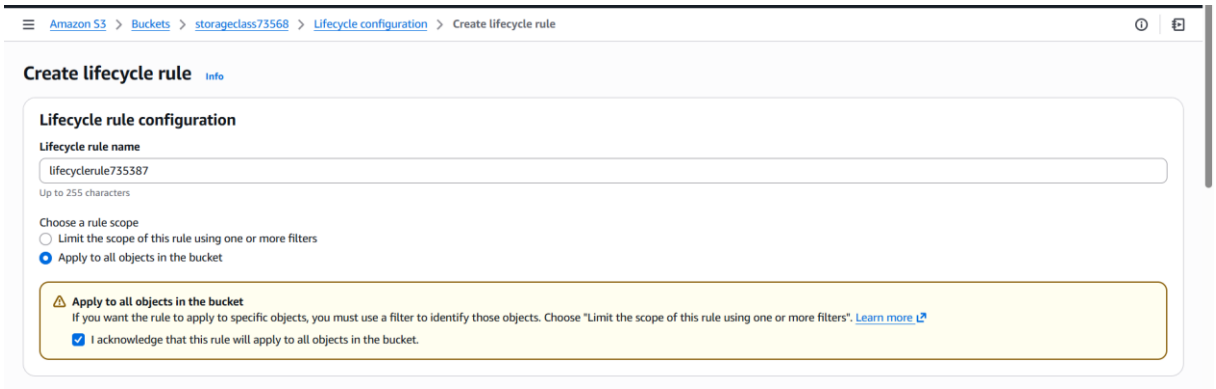
Screenshot 8: Open Management Tab

1. Open the S3 bucket.
2. Click the **Management** tab to access bucket management features.
3. Click **Create lifecycle rule** to configure lifecycle management.



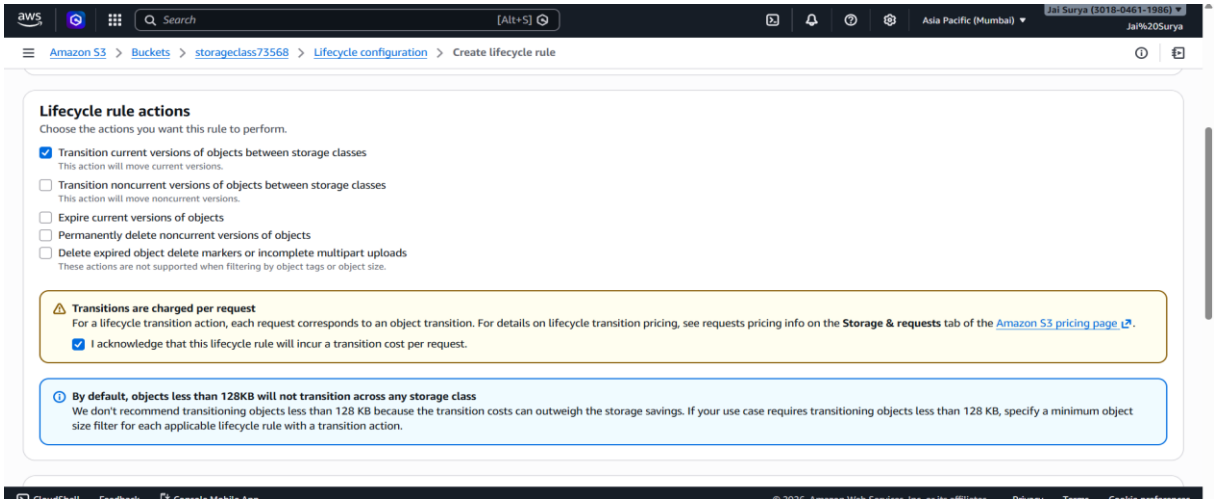
Screenshot 9: Configure Lifecycle Rule

1. Enter a Lifecycle rule name.
2. Select Apply to all objects in the bucket.
3. Proceed to the next step to define lifecycle actions.



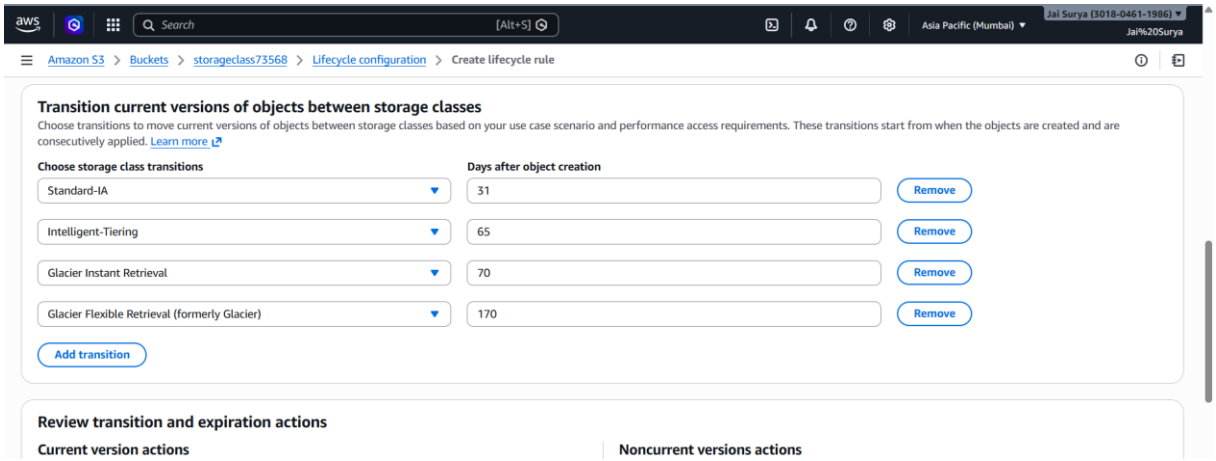
Screenshot 10: Choose Lifecycle Action – Transition Current Version

1. In the lifecycle rule actions, select Transition current versions of objects.
2. Choose the required destination storage class.
3. Proceed to set the transition timing.



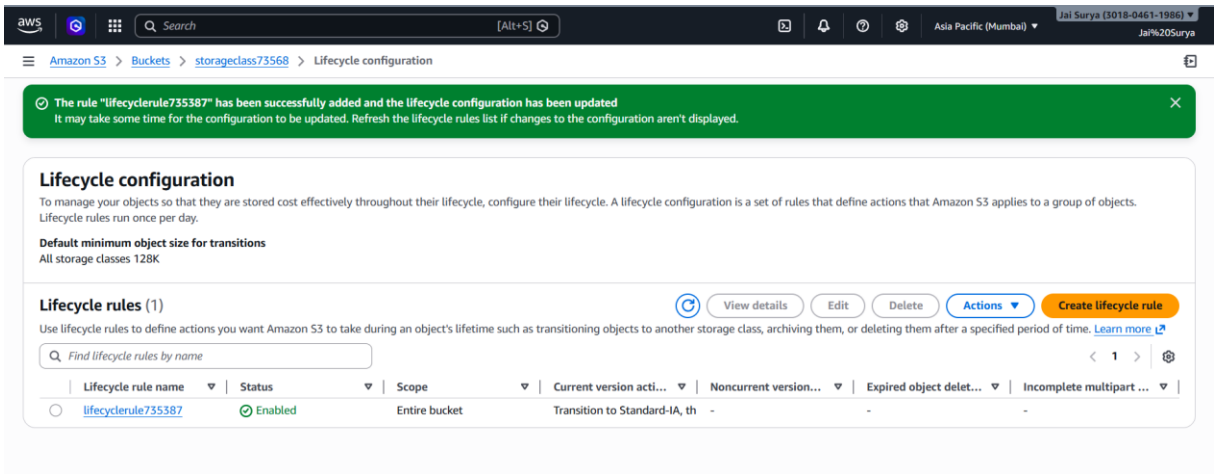
Screenshot 11: Add Transition and Set Timing

1. Click Add transition.
2. Specify the number of days after which objects should transition.



Screenshot 12: Lifecycle Rule Created Successfully

1. Review the lifecycle rule configuration.
2. Click Create rule.
3. The lifecycle rule is created successfully and appears under Lifecycle rules.



S3 SERVER ACCESS LOGGING

Definition

Amazon S3 Server Access Logging records detailed requests made to an S3 bucket, such as object access, upload, download, and deletion. The logs are automatically delivered to a target S3 bucket for analysis and auditing.

Why Server Access Logging Is Used

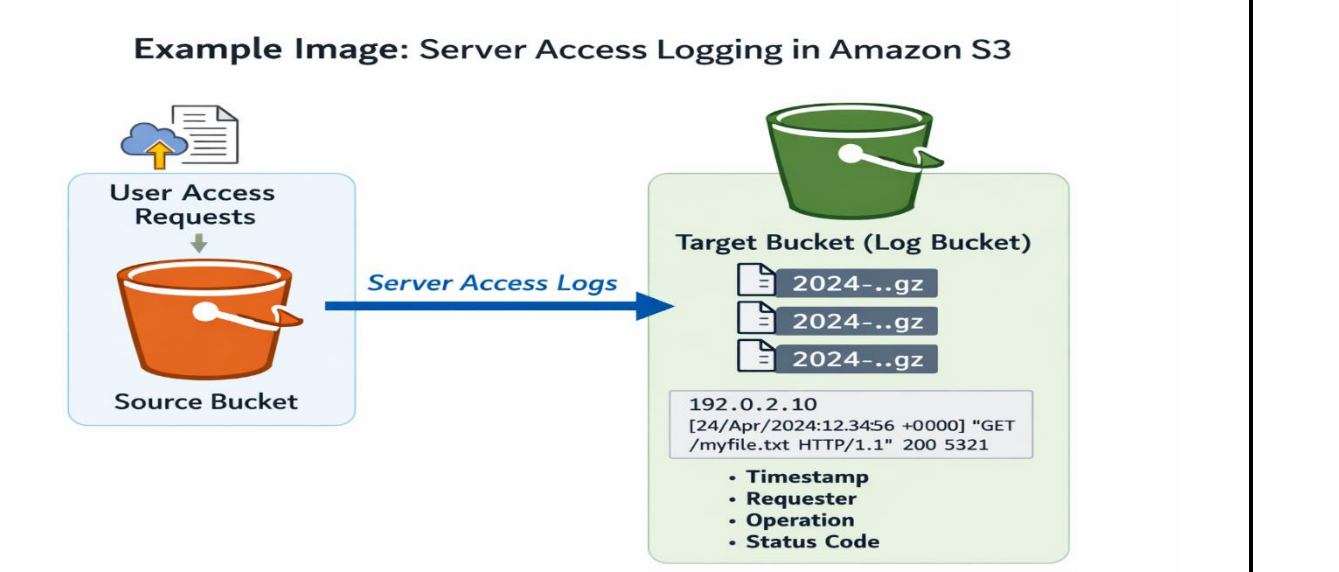
- To monitor bucket access activity
- To track who accessed what and when
- To support security audits and compliance
- To analyze usage patterns and troubleshoot issues

Key Points

- Logs are delivered to a separate destination bucket
- Logging applies to bucket-level requests
- Logs include requester, bucket name, object key, request time, and response status
- Log delivery is not real-time (can take time to appear)

Advantages

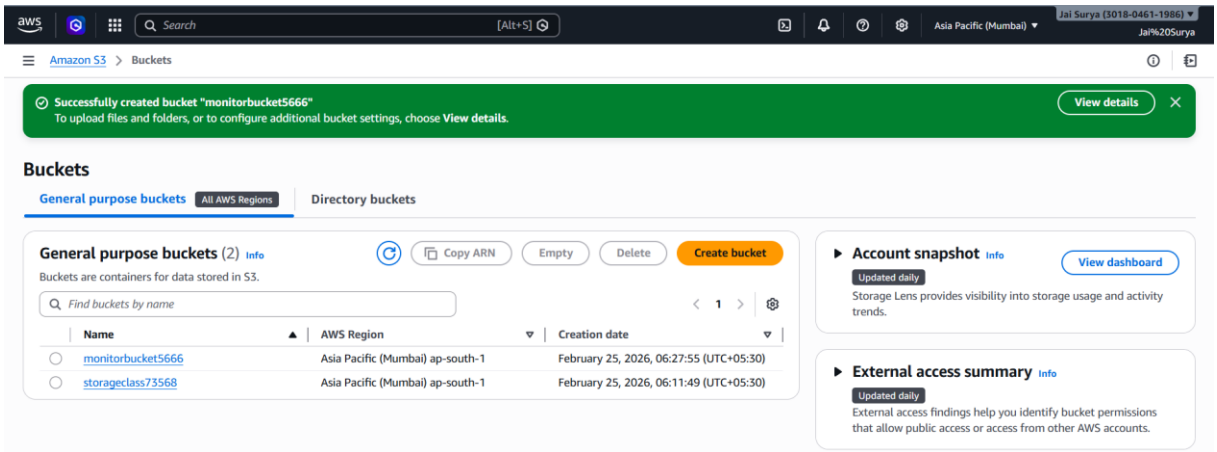
- Improves security visibility
- Helps in forensic analysis
- Useful for compliance and auditing
- No additional configuration required once enabled



HANDS-ON FOR SERVER ACCESS LOGGING

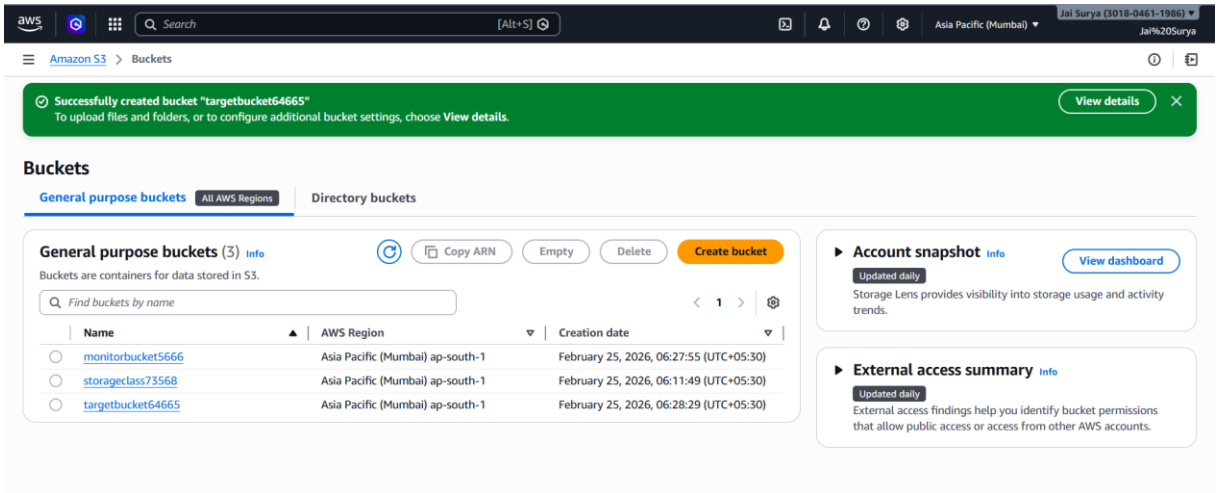
Screenshot 1: Create Source Bucket

1. Open the AWS Management Console.
2. Go to Amazon S3.
3. Click Create bucket and create the source bucket.



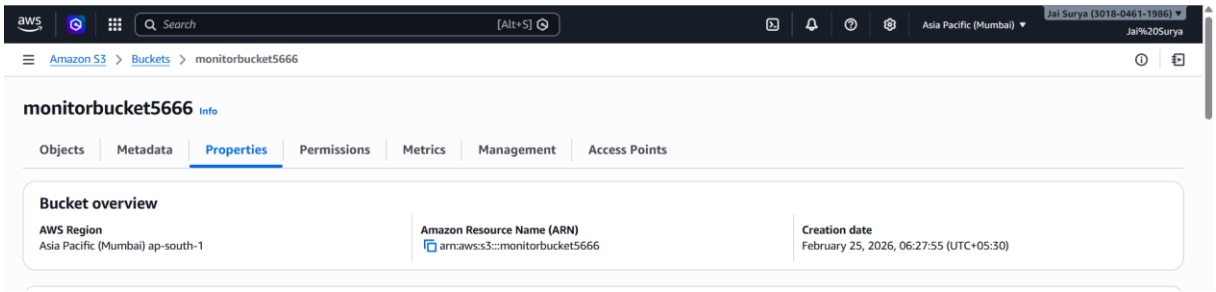
Screenshot 2: Create Destination (Log) Bucket

1. Click Create bucket again.
2. Create a separate bucket to store server access logs.
3. Ensure the bucket is in the same AWS Region



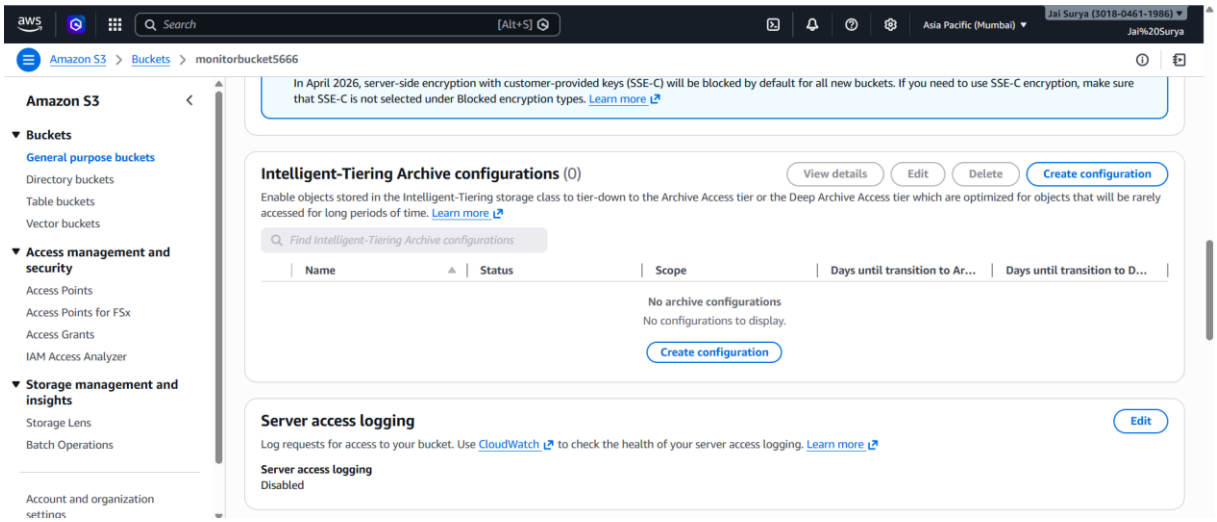
Screenshot 3: Open Source Bucket Properties

1. Open the source bucket.
2. Go to the Properties tab.



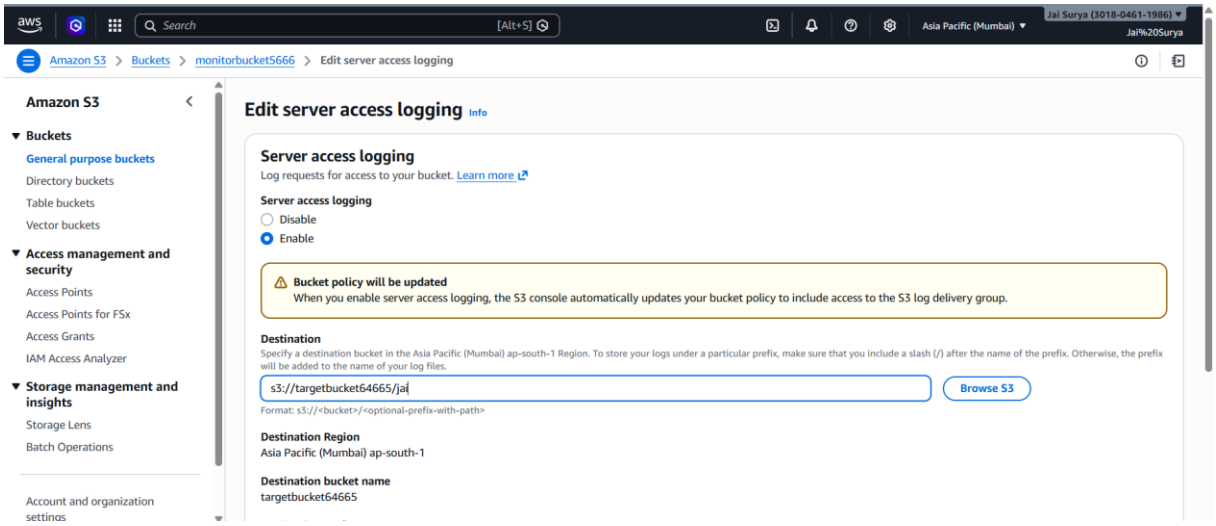
Screenshot 4: Enable Server Access Logging

1. Scroll to Server access logging.
2. Click Edit.



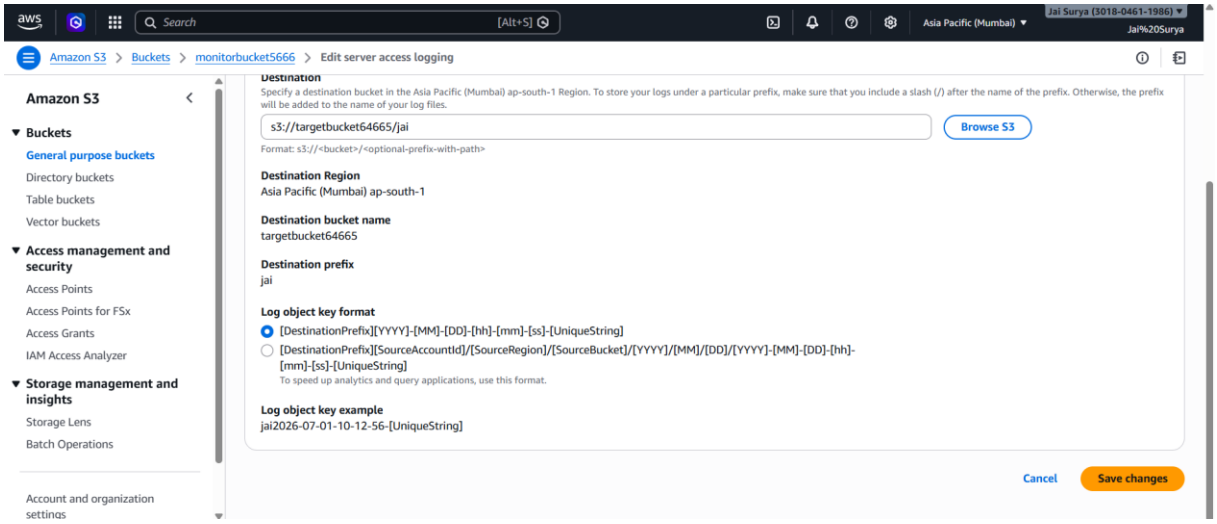
Screenshot 4: Enable Server Access Logging

1. select Enable.
2. Choose the target bucket as the destination.



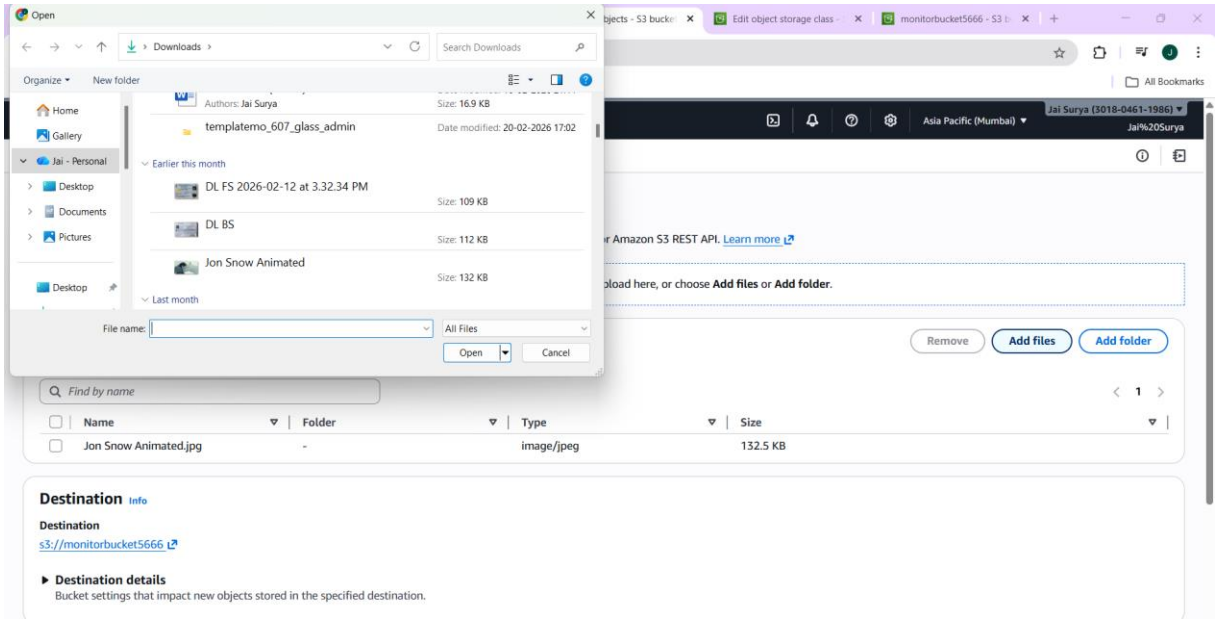
Screenshot 5: Save Logging Configuration

1. Click Save changes.
2. Server access logging is enabled successfully.



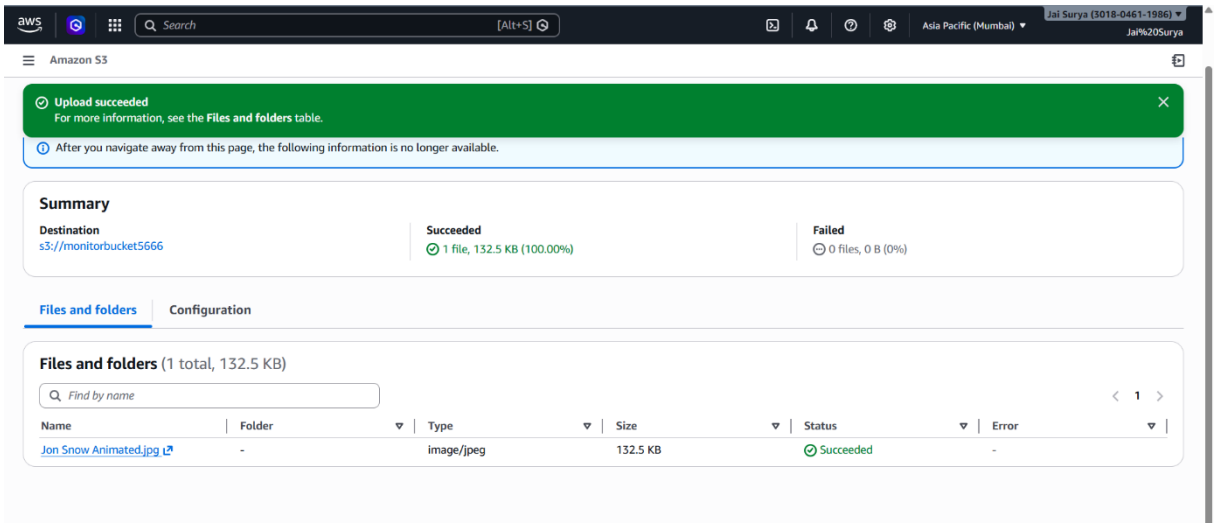
Screenshot 6: Upload File to Source Bucket

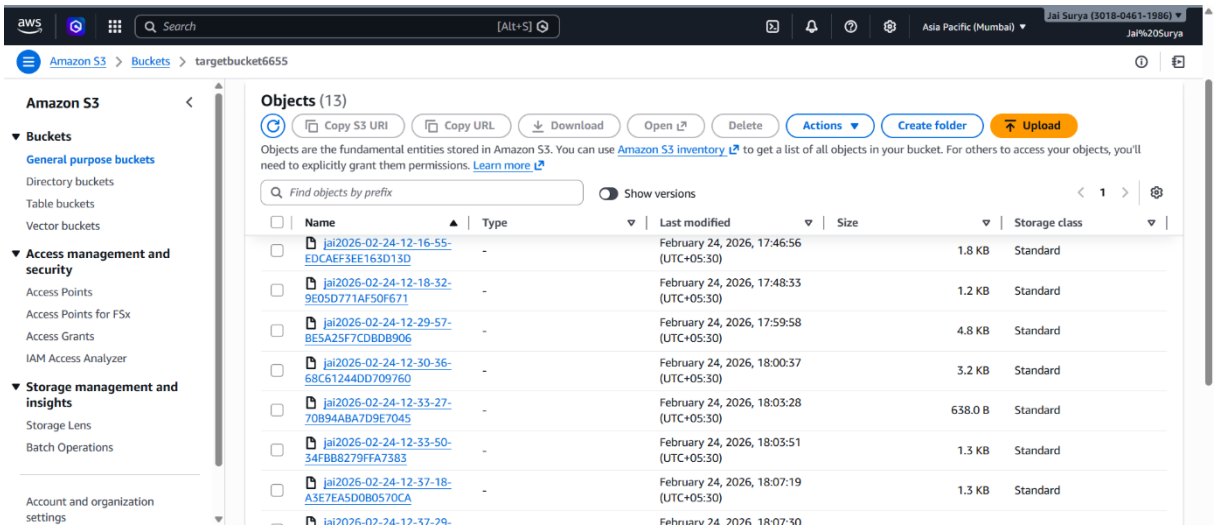
1. Open the source bucket.
2. Click Upload and add a file.



Screenshot: File Uploaded Successfully

1. The selected file is uploaded to the source S3 bucket.
2. Upload status shows successful completion.



Screenshot 7: Logs Updated in Target Bucket

1. Open the target (log) bucket.
2. Server access log files are created successfully.